

15 generative adversarial networks

$$15.1 \mathbb{D}_{\text{JS}}[\bar{P}_r[\bar{x}] \| P_r[\bar{x}]] = \frac{1}{2} \mathbb{D}_{\text{KL}}[\bar{P}_r[\bar{x}] \| P_r[\bar{x}]] + \frac{1}{2} \dots = 0$$

$$15.2 \begin{aligned} L[\Phi] &= - \int P_r[\bar{x}^*] \log \left[\frac{P_r[\bar{x}^*]}{P_r[\bar{x}^*] + P_r[\bar{x}]} \right] dx \\ &\quad - \int P_r[\bar{x}^*] \log \left[\frac{P_r[\bar{x}]}{P_r[\bar{x}] + P_r[\bar{x}^*]} \right] dx \\ &\sim - \int P_r[\bar{x}^*] (\log 2 + \log \left[\frac{P_r[\bar{x}^*]}{P_r[\bar{x}^*] + P_r[\bar{x}]} \right]) dx + \dots \\ &= -2 \left(\frac{1}{2} \int P_r[\bar{x}^*] \log \left[\frac{P_r[\bar{x}^*]}{P_r[\bar{x}^*] + P_r[\bar{x}]} \right] dx \right. \\ &\quad \left. + \frac{1}{2} \int P_r[\bar{x}] \log \left[\frac{P_r[\bar{x}]}{P_r[\bar{x}^*] + P_r[\bar{x}]} \right] dx \right) \\ &= -2 \mathbb{D}_{\text{JS}}[P_r[\bar{x}^*] \| P_r[\bar{x}]] \end{aligned}$$

$$15.3 \mathbb{D}_w(P_r[\bar{x}] \| q[\bar{x}]) = \min_p \left(\sum_{i,j} p_{ij} |i-j| \right)$$

$P \begin{cases} P[\bar{x}] \prod_{i=1,2,3,4} \\ q[\bar{x}] \prod_{i=1,2,3,4} \end{cases}$

In primal form: minimize $\sum p$, such that $A_p = b \wedge p \geq 0$.

$$p = \begin{pmatrix} p_{11} & p_{12} & \dots \\ p_{21} & \dots \\ \dots \end{pmatrix}_{4 \times 4} \quad \left. \vphantom{\begin{pmatrix} p_{11} & p_{12} & \dots \\ p_{21} & \dots \\ \dots \end{pmatrix}} \right\} P_r[\bar{x}=1]$$

$$\begin{pmatrix} 1000 & 1000 & 1000 & 1000 \\ 0100 & 0100 & 0100 & 0100 \\ 0010 & 0010 & 0010 & 0010 \\ 0001 & 0001 & 0001 & 0001 \\ 1111 & 0000 & 0000 & 0000 \\ 0000 & 1111 & 0000 & 0000 \\ 0000 & 0000 & 1111 & 0000 \\ 0000 & 0000 & 0000 & 1111 \end{pmatrix}_{8 \times 16} = \begin{pmatrix} p_{11} \\ p_{21} \\ p_{31} \\ p_{41} \\ p_{12} \\ p_{22} \\ \dots \\ p_{44} \end{pmatrix}_{16 \times 1} = \begin{pmatrix} P_r[\bar{x}=1] \\ P_r[\bar{x}=2] \\ P_r[\bar{x}=3] \\ P_r[\bar{x}=4] \\ q[\bar{x}=1] \\ q[\bar{x}=2] \\ q[\bar{x}=3] \\ q[\bar{x}=4] \end{pmatrix}_{8 \times 1}$$

$$15.4 \quad P[\bar{z}] := \begin{cases} 0 & z < 0 \\ 1 & 0 \leq z \leq 1 \\ 0 & z > 1 \end{cases} \quad \text{vs} \quad q[\bar{z}] = \begin{cases} 0 & z < a \\ 1 & a \leq z \leq a+1 \\ 0 & z > a \end{cases} \quad (a \in [0,1])$$

$$(i) \mathbb{D}_{\text{KL}}[P[\bar{z}] \| q[\bar{z}]] = \int P[\bar{z}] \log \left[\frac{P[\bar{z}]}{q[\bar{z}]} \right] dx \quad \rightarrow \text{not symmetric!}$$

case 1: $a=0$: Then $P[\bar{z}] = q[\bar{z}] \Rightarrow \mathbb{D}_{\text{KL}} = 0$

case 2: $a \neq 0$: Then $\log \left[\frac{P[\bar{z}]}{q[\bar{z}]} \right] = \log \left[\frac{1}{0} \right] \Rightarrow \mathbb{D}_{\text{KL}} = \infty$

$$(ii) \mathbb{D}_{\text{KL}}[q[\bar{z}] \| P[\bar{z}]] = \mathbb{D}_{\text{KL}}[P[\bar{x}] \| q[\bar{x}]] \text{ similarly}$$

$$(iii) \mathbb{D}_{\text{JS}}[P[\bar{x}] \| q[\bar{x}]] = \frac{1}{2} \mathbb{D}_{\text{KL}} \left[P[\bar{x}] \left\| \frac{P[\bar{x}] + q[\bar{x}]}{2} \right\| \right] + \frac{1}{2} \mathbb{D}_{\text{KL}} \left[q[\bar{x}] \left\| \frac{P[\bar{x}] + q[\bar{x}]}{2} \right\| \right]$$

mean divergence of $P[\bar{x}]$ and $q[\bar{x}]$ is the average of the two $\rightarrow$ symmetric!

$P[\bar{z}]$ and $q[\bar{z}]$ overlap in a length of $|a|$, iff $a \leq 1$ (due $P[\bar{z}] = 1$)
For the overlap, $\mathbb{D}_{\text{JS}} = 0$ trivially.

For the non-overlapped part, $\mathbb{D}_{\text{JS}} = \log \left[\frac{1+0}{2} \right]$ always

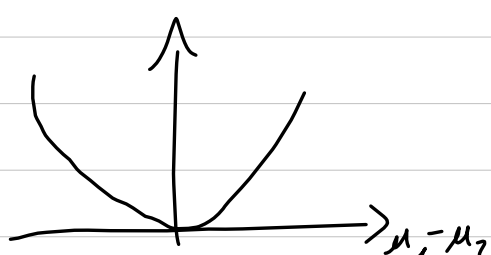
$$\Rightarrow \mathbb{D}_{\text{JS}} = \begin{cases} |a| \log 2 & \text{for } |a| \leq 1 \\ \log 2 & \text{for } |a| > 1 \end{cases}$$

$$(iv) \text{ By similar logic: } \begin{array}{c} | \quad | \quad | \\ a \quad 0 \quad a+1 \end{array}$$

So it's always just a shift: $1 \cdot |a| = |a|$ as emphasis is moved $|a|$ distance

$$15.5 \quad \mathbb{D}_{\text{KL}} = \log \left[\frac{\sigma_2}{\sigma_1} \right] + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2}$$

$$\stackrel{!}{=} \frac{1 + (\mu_1 - \mu_2)^2}{2} - \frac{1}{2} = \frac{(\mu_1 - \mu_2)^2}{2}$$



$$\mathbb{D}_w = (\mu_1 - \mu_2)^2 - 2$$



$$15.6 \quad z \sim \mathcal{N}[0,1]$$

Reject sample earlier than τ SD from 0.

As we have 100 dimensions, and each coordinate

$$X \sim \mathcal{N}(0,1) \text{ has } P[\bar{\tau}] = P[|X| \leq \bar{\tau}] = 2\Phi(\bar{\tau}) - 1,$$

$$P_r[\text{disregarded}] = 1 - (2\Phi(\bar{\tau}) - 1)^{100}$$

$$(i) \tau = 2.0 \Rightarrow P_r[\text{disregarded}] = 99.05\%$$

$$(ii) \tau = 1.0 \Rightarrow P_r[\text{disregarded}] = 99.99\%$$

$$(iii) \tau = 0.5 \Rightarrow P_r[\text{disregarded}] = 1$$

$$(iv) \tau = 0.04 \Rightarrow P_r[\text{disregarded}] \approx 0$$

⚡ COD